



1 & 4. INFRASTRUCTURE

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2. CEMENT The per capita consumption in India is 257kg, far less than the global average of 540kg

3. STEEL The companies are increasing use of scrap and piloting production of green hydrogen to reduce carbon footprint



1. PHOTO BY ANI; 2&4. PHOTO BY HARDIK CHHABRA

Almost 70% of the infrastructure that India needs to become developed by 2047, whether it's roads, ports, airports or power plants, is yet to be built, leaving huge scope for the country to make its public works sustainable.

With that in mind, steel and cement makers in the world's fifth-largest economy are transitioning towards low-carbon manufacturing by reducing dependence on coal to fire their plants. They are investing in new energy-efficient technolo-

gies, switching to renewable energy and even using waterways and trucks powered by liquefied natural gas and electricity. Steel makers are increasing the use of scrap and piloting production of green hydrogen.

Of course, there's a flip side to the effort. Integrating sustainability means spending more money. According to a steel ministry assessment, conventional steel costs ₹55,000 per metric tonne, while green steel with 10% blending costs ₹60,500; 20% and 30% blended products cost ₹66,000 and ₹71,500 per MT, respectively.

The effort is not limited to manufacturing plants. For instance, Carbon Bank, launched by Tata Steel last year, helps customers adopt low-carbon steel aligned with global standards. Jindal Stainless recently inaugurated India's first green hydrogen plant in the stainless steel industry to cut annual carbon emissions by 2,700 tonnes. ArcelorMittal Nippon Steel India is targeting about 70% output through green technology by 2027.

Cement companies, too, are pumping large investments into renewable power and sustainable